COSC 561

Project 1

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1 Part A: Plan a delivery with A*

1.1 Formulation

The formulation of the problem is as follows:

1. State: A tuple $(row, column)$ representing the robot's current position.
2. Initial State: The robot's starting position, which must correspond to a selected building/building entrance.
3. Actions: UP, $DOWN$, $LEFT$, and $RIGHT$
4. Transition model: An action attempts to move the robot one cell in the corresponding direction. If the destination is blocked or outside the map, the robot remains in place.
5. Valid states: Any traversable cell, including . and buildings (S, L, H, D, P, W).
6. Invalid states: Non-traversable cells, obstacles denoted by #.
7. Path cost: One unit per action. Total cost is the number of actions taken.
8. Goal test: The robot's current state equals the selected destination's coordinates.

Breadth-first search is guaranteed to find the shortest path as when compared with depth-first search because every movement/action costs 1 and BFS explores states/nodes in order of increasing distance from the start. DFS *could* also find the shortest path, but it has a chance to find a longer more roundabout route first. A* itself can also be led down certain corridors such that BFS will find the shortest path with fewer expansions first, but with the Manhattan distance as $h(s)$ the shortest path will still ultimately be found.

1.2 Regarding manhattan(state, goal)

Manhattan distance is admissible as a heuristic here because movement is in the four cardinal directions. The shortest distance to the goal cannot be overestimated.

1.3 Running BFS and A*

As was discussed previously in Section 1.2, because the Manhattan Distance is admissible for our unit-cost graph, A* will always find the shortest path, just like BFS. $h(s)$ favors these graphs well: all A* expansion counts are fewer than the BFS expansions (and by considerable margins in some cases).

Table 1: BFS and A* Search Results

Map	**Task**	**Route Length**		**Expanded Nodes**	
		BFS	**A***	**BFS**	**A***
Baseline	S → W	18	18	67	22
Baseline	L → D	22	22	79	62
Baseline	H → P	16	16	41	16
Detour	S → W	26	26	70	51
Detour	L → D	22	22	65	60
Detour	H → P	16	16	41	16

2 Part B: Learn a delivery policy

2.1 Main Observations

Q-learning enabled all 9 agents (3 per task) to independently learn shortest-path policies without access to the transition model. Results were consistent across seeds, with no variations in success rate, mean delivery steps, or mean return. Right off the bat this tells us the 2000 episodes was sufficient for training. Because there were no failed deliveries during evaluation, the reward is simply $40 - step_count$. A summary of the 3 tasks and how the agent results compared to BFS and A* can be found in 2.

Table 2: Q-Learning Evaluation Compared with BFS and A*

Task	Success Rate	Mean Steps	Mean Return	BFS Steps	A* Steps
S → W	100%	18	22	18	18
L → D	100%	22	18	22	22
H → P	100%	16	24	16	16

2.2 Additional Observations and Plots

I've picked two metrics to focus on: training success and learning efficiency.

2.2.1 Training Success

1. L→ D reaching 100% training success rate by episode 30
2. H→P reaches 100% by episode 50 despite having the shortest optimal path (16).
3. S→W reaches 100% the slowest, reaching 100% by episode 60, with the greatest variability between seeds during training

I've plotted training success using 10 episode blocks in 1. 10 as the block size was selected to better visualize the curves, 'smoothing out' noise/little variations. The mean is plotted along with the area corresponding to one standard deviation for each task. It's clear that the 2000 episodes were overkill for these tasks, but as we'll see, the convergence iterations for success and minimum delivery steps differ.

2.2.2 Learning Efficiency

1. All three tasks demonstrate a dramatic reduction in steps in the first 100 to 150 episodes.
2. The mean steps settle around 20 for H→P, 23 for S→W, and 28 for L→D

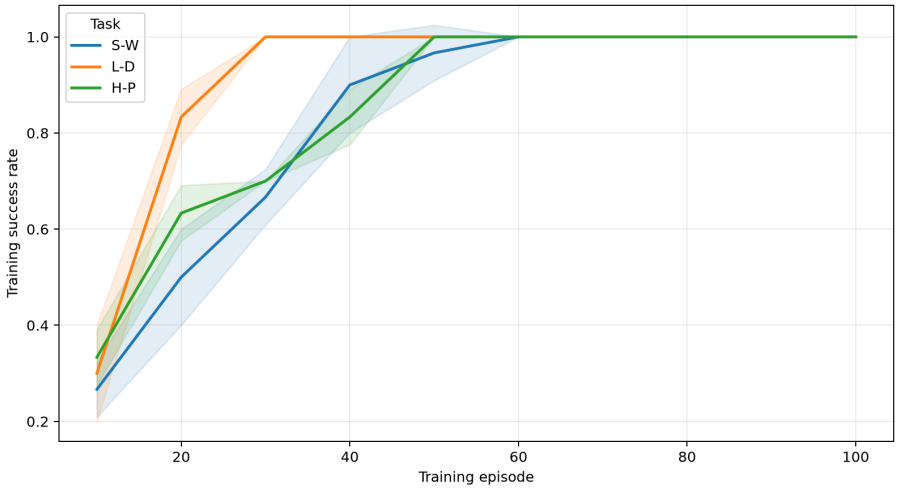


Figure 1: Training convergence for each task by mean success rate using 10 episode blocks.

3. All three tasks remain above their optimal shortest-path lengths, along with jitter and the occasional spike in standard deviation. This is because $\epsilon = 0.2$ remains constant throughout training. The agent will continue exploring even after learning a successful policy.

I've plotted mean steps per successful training episode, using blocks of 25 2. For this example, I decided to show the entire 2000 episode domain to demonstrate that because ϵ is constant, during training the mean number of the steps for each task will not converge to the optimal value.

3 Part C: Designing My Own Map

3.1 Map Design

I created my very own map, 'Welby Land' 3. I wanted to see how Q-learning handled the narrow, winding corridors of a glorious 'W'. S and W are positioned at the upper vertices, requiring a long traversal through the W to complete delivery. The map includes loops and alternative routes along the lower corridors, as well as dead-end branches such as Putnam Hall (P). These features make it useful for investigating how discount factors affect learning over long routes.

3.2 BFS and A* Results

See table 3. Optimal shortest-path is 52.

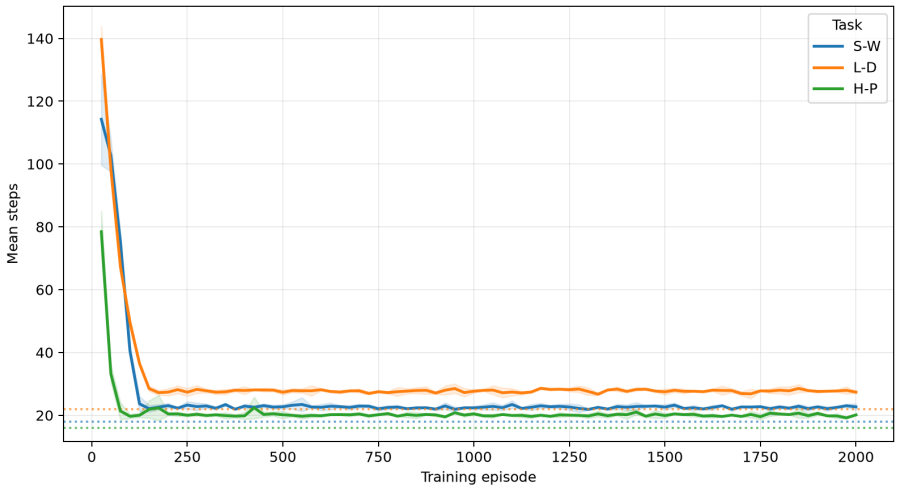


Figure 2: mean steps per successful training episode, using 25 episode blocks. Dotted lines represent optimal number of steps.

```
#########################
##S###################W##
##..##################.##
###.##################.##
###.########P#########..##
###..########.#########.###
####.########.#########.###
####.########H#########.###
####.########..#######..###
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#######....#######.#.#######
########..########...#######
########..#########..#######
########L#########D########
#########################
```

Figure 3: Welby Land, replete with winding, narrow corridors.

Table 3: BFS and A* Search Results for Welby Land

Algorithm	**Path Length**	**Expanded Nodes**
BFS	52	90
A*	52	69

3.3 Research Question and Prediction

Research question: *How do discount factors affect Q-learning's ability to propagate reward information backward through long routes?* **Prediction:** *Lower discount factors will slow learning or produce worse policies.*

3.4 Experimental Methodology

Using the suggested discount factors of $\gamma = 0.5, 0.8, 0.95$, I conducted three experiments on Welby Land, each using independent seeds 0, 1, and 2. All other parameters were held constant as per the provided defaults: 2000 training episodes, 100 evaluation episodes per seed, learning rate $\alpha = 0.3$, exploration rate $\epsilon = 0.2$, slip probability $p = 0$, and a maximum of 200 steps per episode. Each agent was trained with epsilon-greedy exploration and evaluated using its learned policy with $\epsilon = 0$. Training success rates, delivery steps, and final evaluation performance were compared across the three discount factors.

3.5 Results and Discussion

The results partially support my prediction that lower discount factors would slow learning or produce worse policies. As shown in Figure 4, $\gamma = 0.8$ and $\gamma = 0.95$ reached sustained 95% training success by episode 200, while $\gamma = 0.5$ required 250 episodes.

Figure 5 shows a similar pattern. All three configurations initially struggled, with mean returns near -200, before improving rapidly between episodes 100 and 300. Higher discount factors generally improved faster, although the differences diminished as training progressed.

As shown in Figure 6, all three configurations eventually stabilized around 65–67 steps per successful training episode, above the optimal 52-step route found by BFS and A*. This is consistent with the constant exploration rate of $\epsilon = 0.2$, which continues introducing fun little random actions during training.

Despite the differences in learning speed, all nine agents achieved 100% evaluation success and optimal 52-step routes, producing a mean return of -12. Thus, lower discount factors appeared to slow initial learning but did not produce worse final policies, at least with the given experimental parameters.

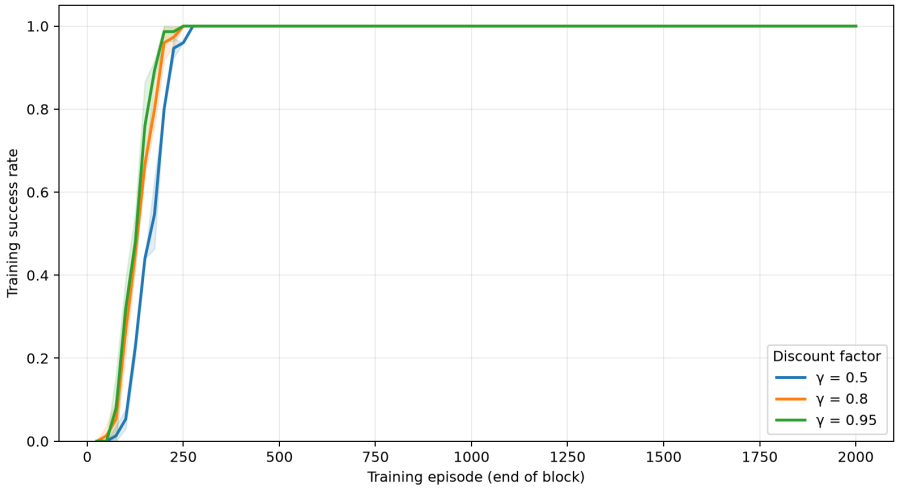


Figure 4: Training success rates across discount factors using 25-episode blocks.

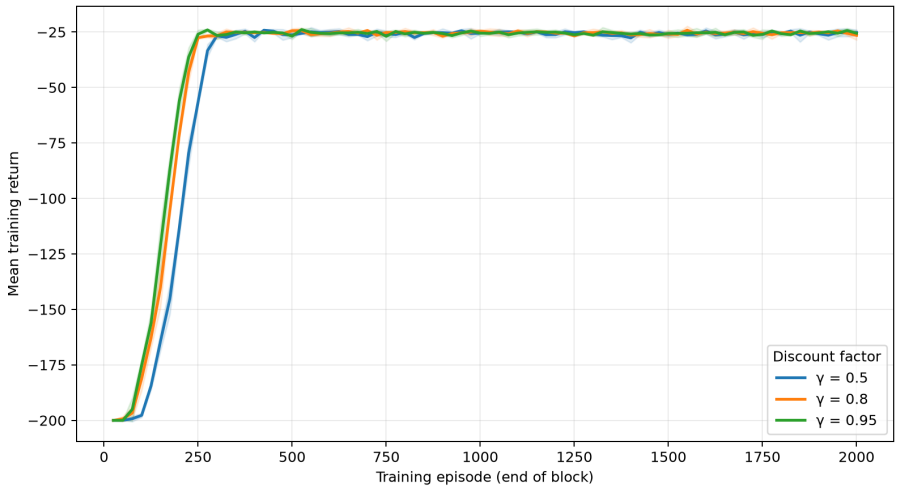


Figure 5: Mean training returns across discount factors using 25-episode blocks.

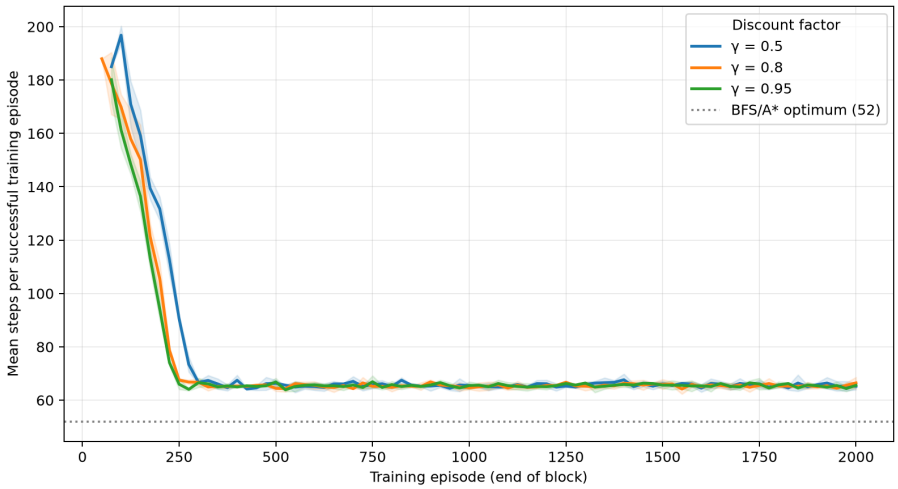


Figure 6: Mean steps per successful training episode across discount factors. 25-episode blocks. The dotted line represents the optimal 52-step route.